

COS 433: Cryptography

Announcements:

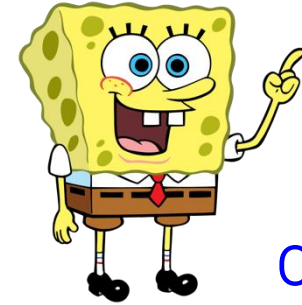
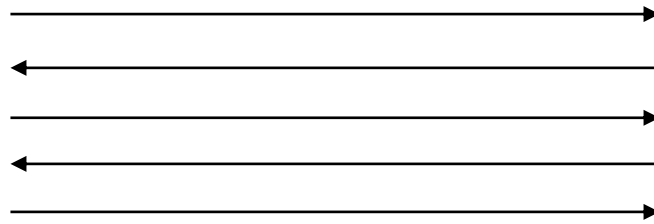
- Problem Set 5 due 12/3
- **Final problem set** released 12/1, due **Sunday 12/14 11:59pm**
 - No collaboration allowed
 - You may use anything available on our course website (including the lecture slides and textbooks)
 - No other sources may be consulted
 - No late submissions (due date assigned by the university)

Proof Systems

Claim: $x \in L$



$P(x, w)$



$V(x)$

Output: $b \in \{0,1\}$

Definition: a proof system for a language $L \subset \{0,1\}^*$ consists of a pair of interactive algorithms (P, V) run on an input $x \in \{0,1\}^n$

- V will always be a **poly**(n)-time algorithm
- P can be inefficient, or **poly**(n)-time given a witness w

It must satisfy the following properties:

1. **Completeness:** if $x \in L$ and the two parties run (P, V) , then V outputs 1 (accept)
2. **Soundness:** if $x \notin L$, then for **any** strategy P^* , V accepts with probability **negl**(n).

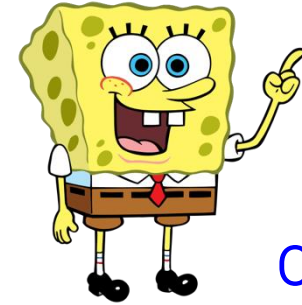
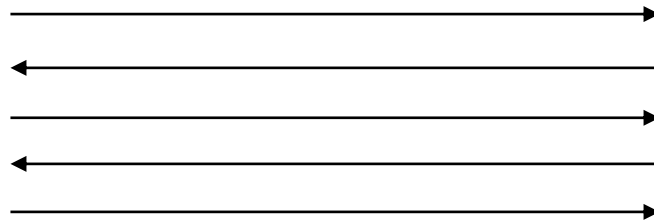
Zero-Knowledge Proofs

[Goldwasser-Micali-Rackoff '85]

Claim: $x \in L$



$P(x, w)$



$V(x)$

Output: $b \in \{0,1\}$

Definition: a proof system (P, V) for a language L is malicious verifier zero knowledge if for all polynomial-time V^* , there exists a polynomial-time simulator

$$\text{Sim}^*(x) \rightarrow \tau$$

such that for all $x \in L$ and all valid w ,

$$\text{Sim}^*(x) \approx_c \text{view}_{V^*}(x, w)$$

Simulation paradigm: everything V^* learned, V^* could have generated on their own!

Zero-Knowledge Proofs

Theorem (last class):

The language L_{QR} has a **perfect zero knowledge** (PZK) proof system.

- PZK: Simulation indistinguishability is perfect

Question: for which languages are PZK proofs possible?

- Factoring: unknown (may also depend on the formulation of the language)
- 3-coloring: unknown but **strong evidence against**

Next: computational zero knowledge



Zero-Knowledge Proofs for NP



Theorem [Goldreich-Micali-Wigderson]:

Assuming OWFs, every language $L \in \mathbf{NP}$ has a ZK proof system!

- Honest prover efficient given $\mathbf{NP}$ -witness.
- ZK is computational



All languages with (non-interactive) proofs have ZK proofs!

Can be extended to all languages with interactive proofs ($\mathbf{IP}$)

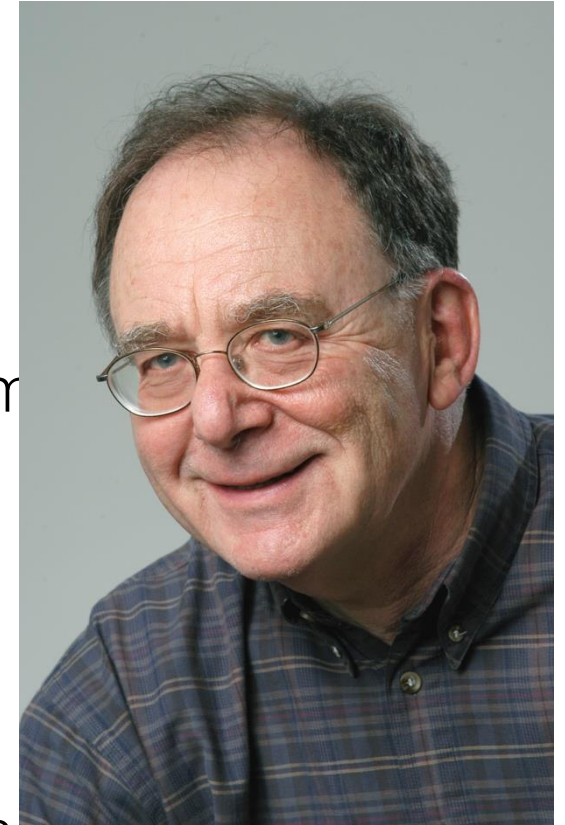
Zero-Knowledge Proofs for NP



Wigderson



of system



Proc

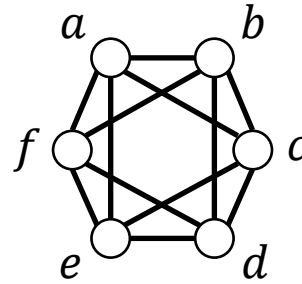
Step 1. it suffices to build a ZK proof system for $L_{3\text{Col}}$ (3-colorability)

Makes use of “NP-completeness:” every **NP** problem can be efficiently converted into an instance of 3-coloring! ([Cook 1971], [Levin 1972], [Karp 1972])

Zero Knowledge Proofs for NP

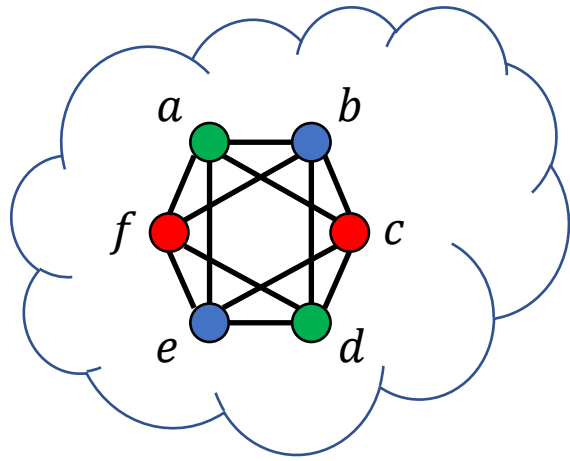
Claim: This graph
has a 3-coloring.

P



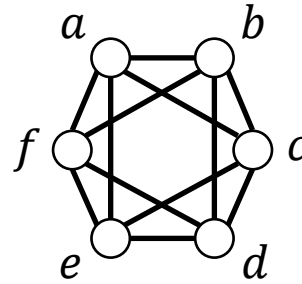
V

Zero Knowledge Proofs for NP



Claim: This graph
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... P

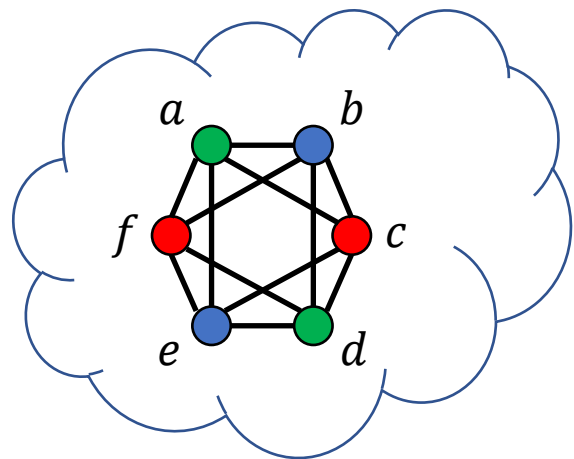


V

Intuition:

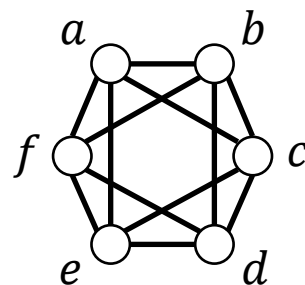
- P will send a message defining $|E(G)|$ different claims
- V will test one of these claims at random

Zero Knowledge Proofs for NP



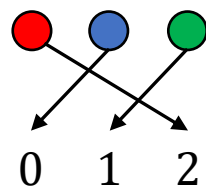
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○ ○ ○ P



V

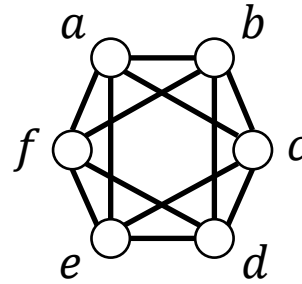
1) Randomize colors:



Zero Knowledge Proofs for NP

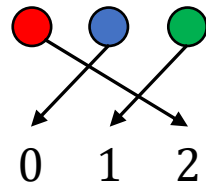
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V

1) Randomize colors:



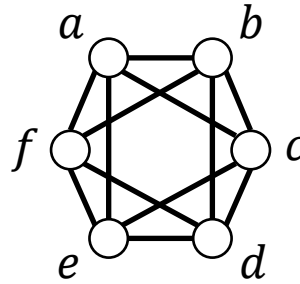
2) Commit:

a	b	c	d	e	f
1	0	2	1	0	2

Zero Knowledge Proofs for NP

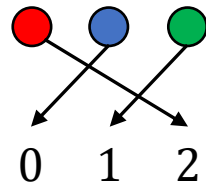
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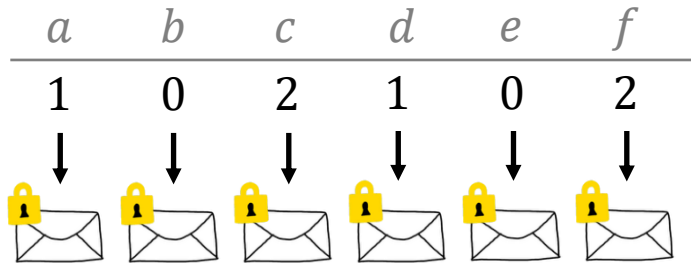


V

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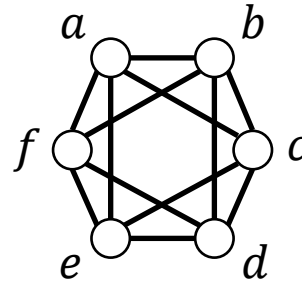


2) Commit:



Zero Knowledge Proofs for NP

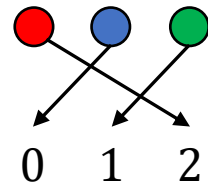
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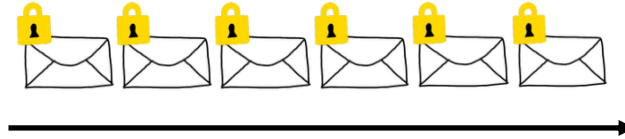
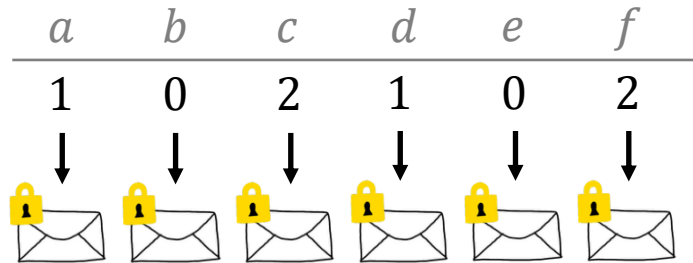
P

V

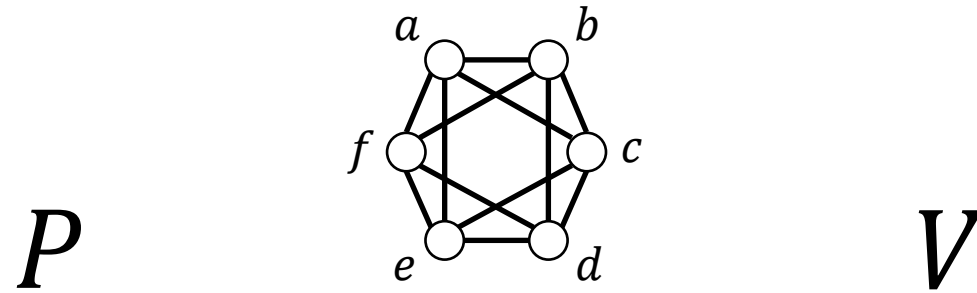
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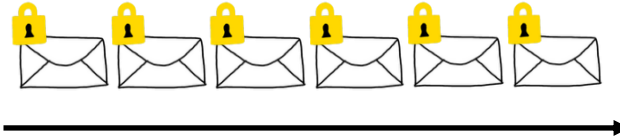
2) Commit:



Zero Knowledge Proofs for NP

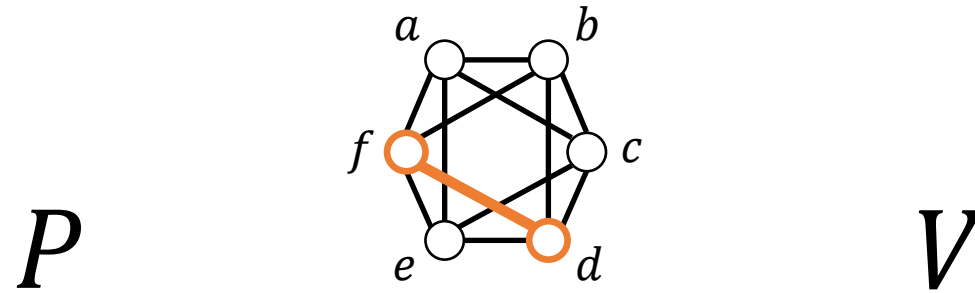


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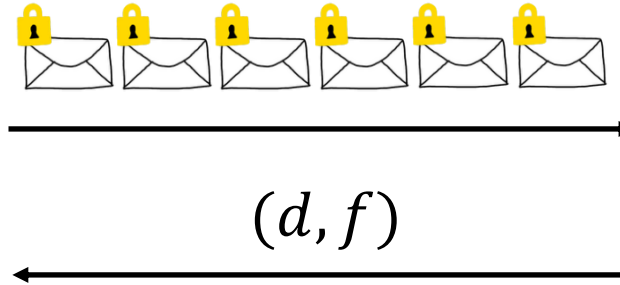
2) Commit:

Zero Knowledge Proofs for NP



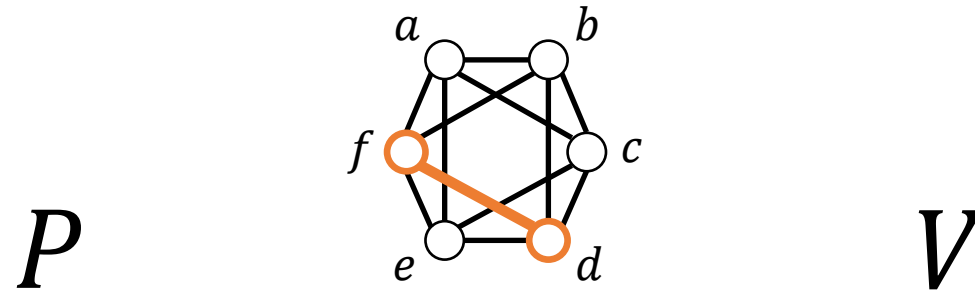
1) Randomize colors:

2) Commit:



1) Sample a **challenge** edge.

Zero Knowledge Proofs for NP



1) Randomize colors:



2) Commit:

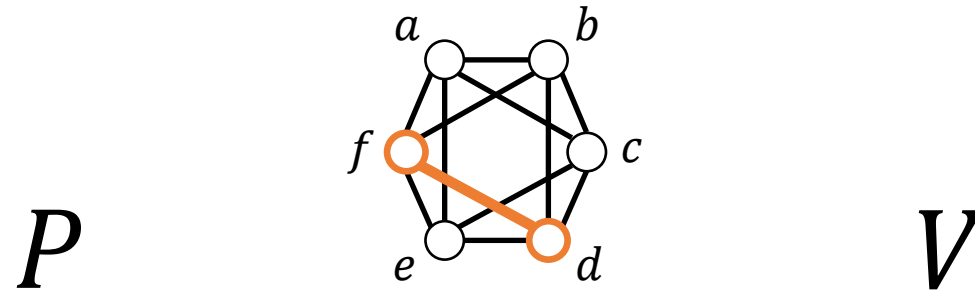
(d, f)

1) Sample a **challenge** edge.

3) Reveal edge colors



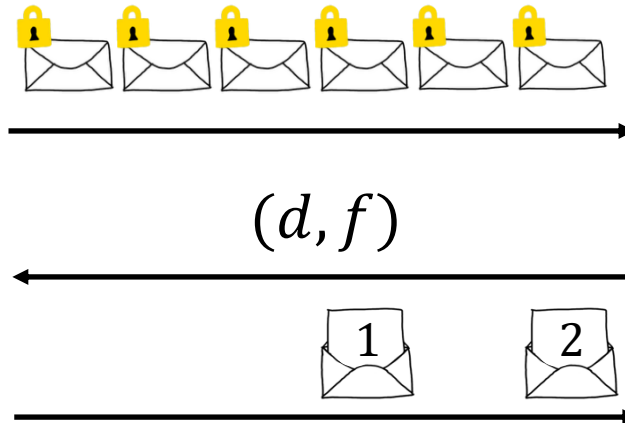
Zero Knowledge Proofs for NP



1) Randomize colors:

2) Commit:

3) Reveal edge colors



1) Sample a **challenge** edge.

2) Accept if colors are different.

Tool: Commitment Schemes

Commitment Schemes

Definition: a (2-message) commitment scheme consists of an efficient algorithm

$$\text{Com}(\text{ck}, m; r)$$

satisfying the following two properties:

1. Computational Hiding: **for all** $\text{ck} \in \{0,1\}^\lambda$ and all pairs of messages $m_0 \neq m_1$:

$$\text{Com}(\text{ck}, m_0) \approx_c \text{Com}(\text{ck}, m_1)$$

2. Statistical Binding: with probability $1 - \text{negl}(\lambda)$ over the choice of $\text{ck} \leftarrow \{0,1\}^{\ell(\lambda)}$, there **do not exist** $m_0 \neq m_1$ and r_0, r_1 such that $\text{Com}(\text{ck}, m_0; r_0) = \text{Com}(\text{ck}, m_1; r_1)$

Commitment Schemes

Intuition: a (2-message) commitment scheme consists of an efficient algorithm

$$\text{Com}(\text{ck}, m; r)$$

satisfying the following two properties:

1. **Computational Hiding:** $\text{Com}(\text{ck}, m)$ does not reveal m (commitment key is **public**)
2. **Statistical Binding:** commitment c is only consistent with a single m (known **ck**)
 - Terminology: **opening** a commitment (for us) means revealing the randomness

Commitment Schemes

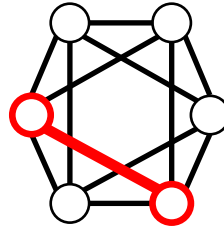
Theorem: if PRGs exist, then commitment schemes exist!

Construction (analysis on board):

$$\text{Com}(\text{ck} \in \{0,1\}^{3\lambda}, m; \textcolor{red}{r} \in \{0,1\}^\lambda) = G(\textcolor{red}{r}) \oplus m \cdot \text{ck}$$

Zero Knowledge Proofs for NP

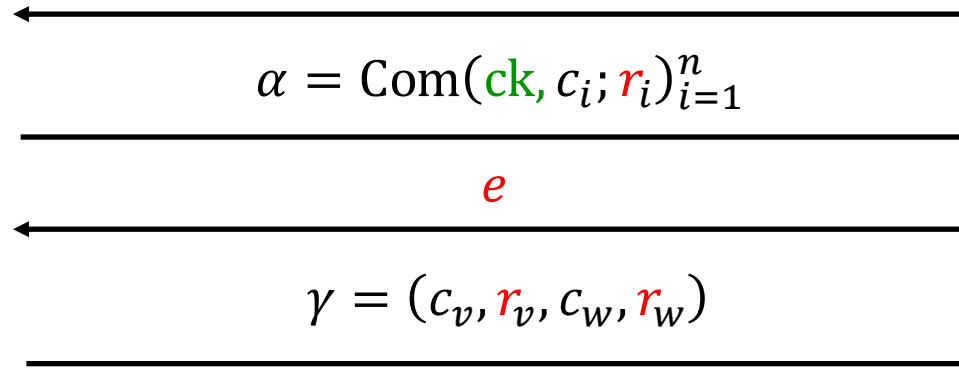
P



V

Sample $\pi \leftarrow S_3$
 Define $c_i = \pi \circ \chi(i)$
 Sample r_i for every i

Sample ck



$e = (v, w) \leftarrow E(G)$

Check:

- $\alpha_v = \text{Com}(c_v; r_v)$
- $\alpha_w = \text{Com}(c_w; r_w)$
- $c_v \neq c_w$

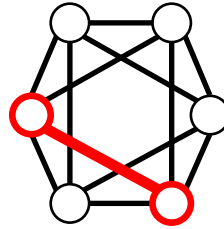
Completeness: π preserves proper colorings

Soundness: if G is **not** 3-colorable:

- (for almost all ck) Every string α implicitly defines a fixed coloring χ^* of G (by **binding** of Com)
- **There exists an edge** $e^* = (v, w)$ such that $\chi^*(v) = \chi^*(w)$
- If $e = e^*$ (inverse poly chance), P^* is stuck! (Repeat protocol to improve soundness error)

Zero Knowledge Proofs for NP

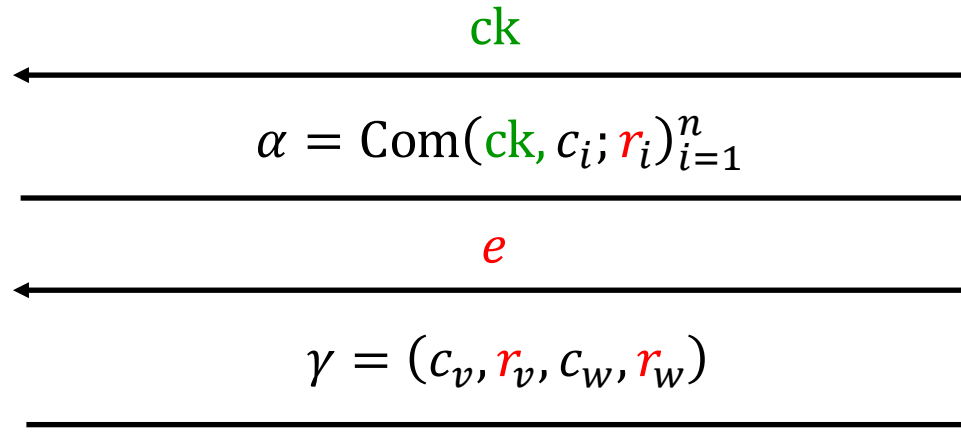
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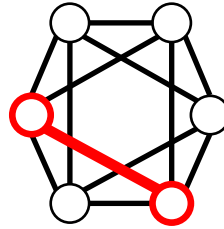
- $\alpha_v = \text{Com}(c_v; r_v)$
- $\alpha_w = \text{Com}(c_w; r_w)$
- $c_v \neq c_w$

ZK Intuition: Verifier sees

- Two random unequal colors
- Commitments to those two colors
- Unintelligible commitments to $n - 2$ other colors

Zero Knowledge Proofs for NP

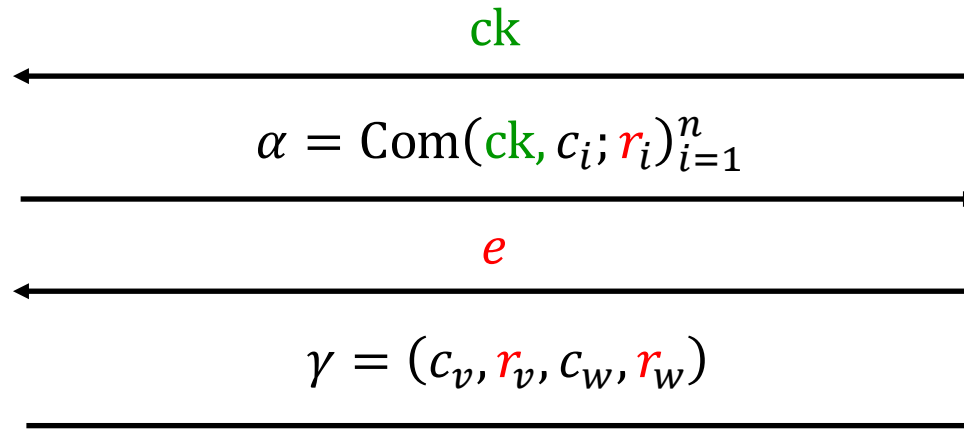
Sim



V

Sample $e \leftarrow E(G)$
 Sample $c_v \neq c_w$
 Set $c_i = 0$ for other i
 Sample r_i for every i

Sample ck



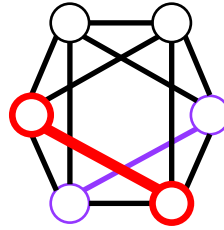
HVZK:

For 3-colorable G , this is like $\text{view}_V(x, w)$, but $c_i = \pi \circ \chi(i)$ has been replaced by 0 (for $i \notin \{v, w\}$)

$\text{Sim}(x) \approx_c \text{view}_V(x, w)$ by the hiding property of Com

Zero Knowledge Proofs for NP

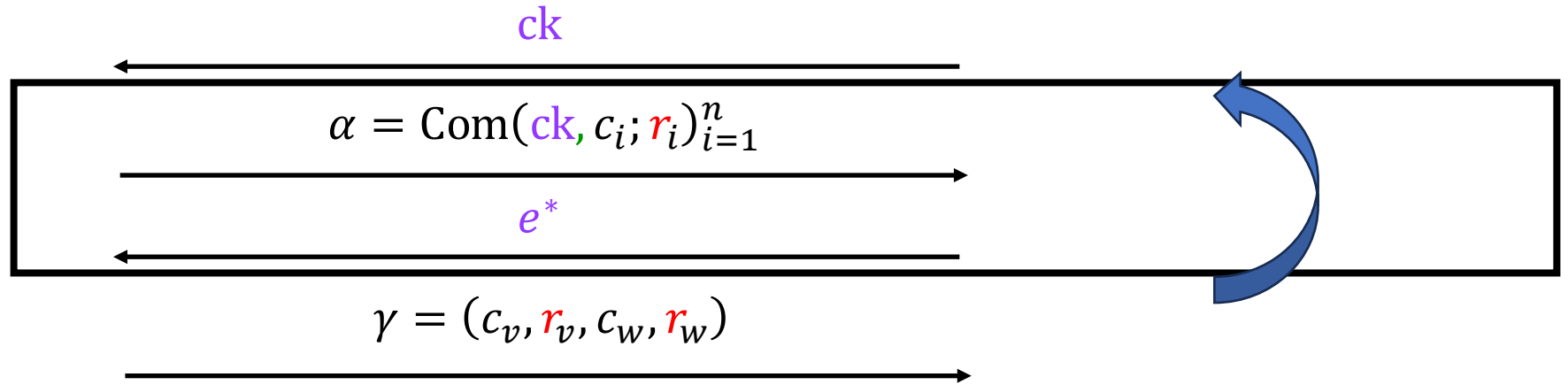
Sim^*



V^*

Sample $e \leftarrow E(G)$
 Sample $c_v \neq c_w$
 Set $c_i = 0$ for other i
 Sample r_i for every i

If $e^* = e$:
 Else, "rewind"



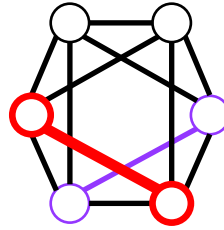
ZK: same idea as for QR

Claim 1: terminates in $|E(G)| \cdot \lambda$ attempts with all but negligible probability

Proof: by Com hiding, $\Pr[e^* = e] = \frac{1}{|E(G)|} + \text{negl}(\lambda)$ (else V^* has learned info about c)

Zero Knowledge Proofs for NP

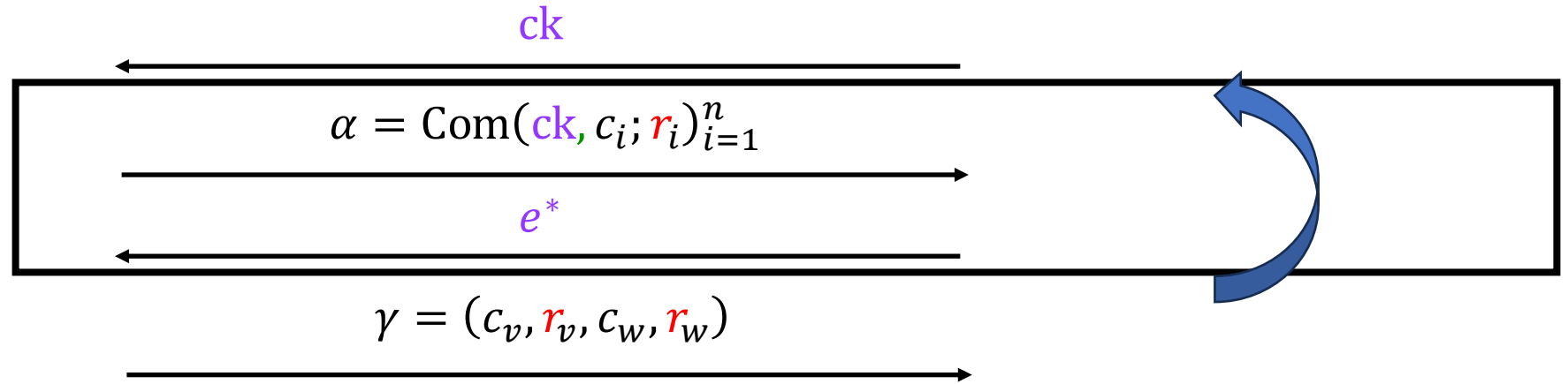
Sim^*



V^*

Sample $e \leftarrow E(G)$
Sample $c_v \neq c_w$
Set $c_i = 0$ for other i
Sample r_i for every i

If $e^* = e$:
Else, "rewind"



ZK: same idea as for QR

Claim 1: terminates in $|E(G)| \cdot \lambda$ attempts with all but negligible probability

Claim 2: output computationally indistinguishable from $\text{view}_{V^*}(G, \chi)$ (by Com hiding)

Zero-Knowledge Proofs for NP

Theorem [Goldreich-Micali-Wigderson]:

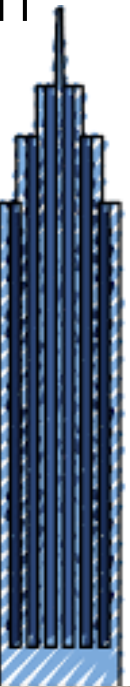
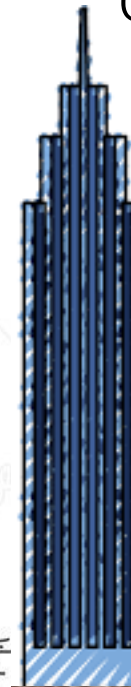
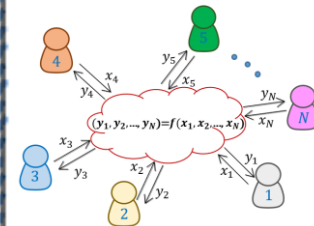
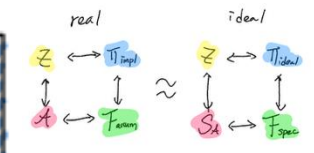
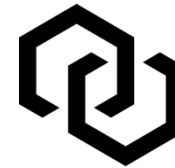
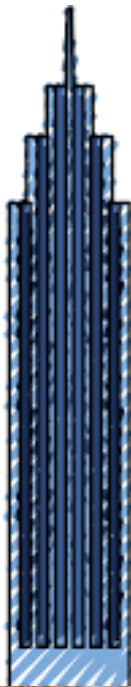
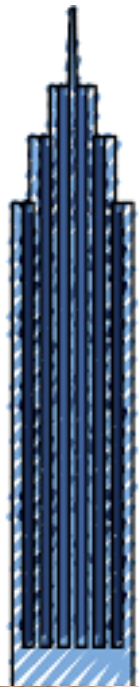
Assuming OWFs, every language $L \in \mathbf{NP}$ has a ZK proof system!

Proof Systems as a Foundation

digital signatures

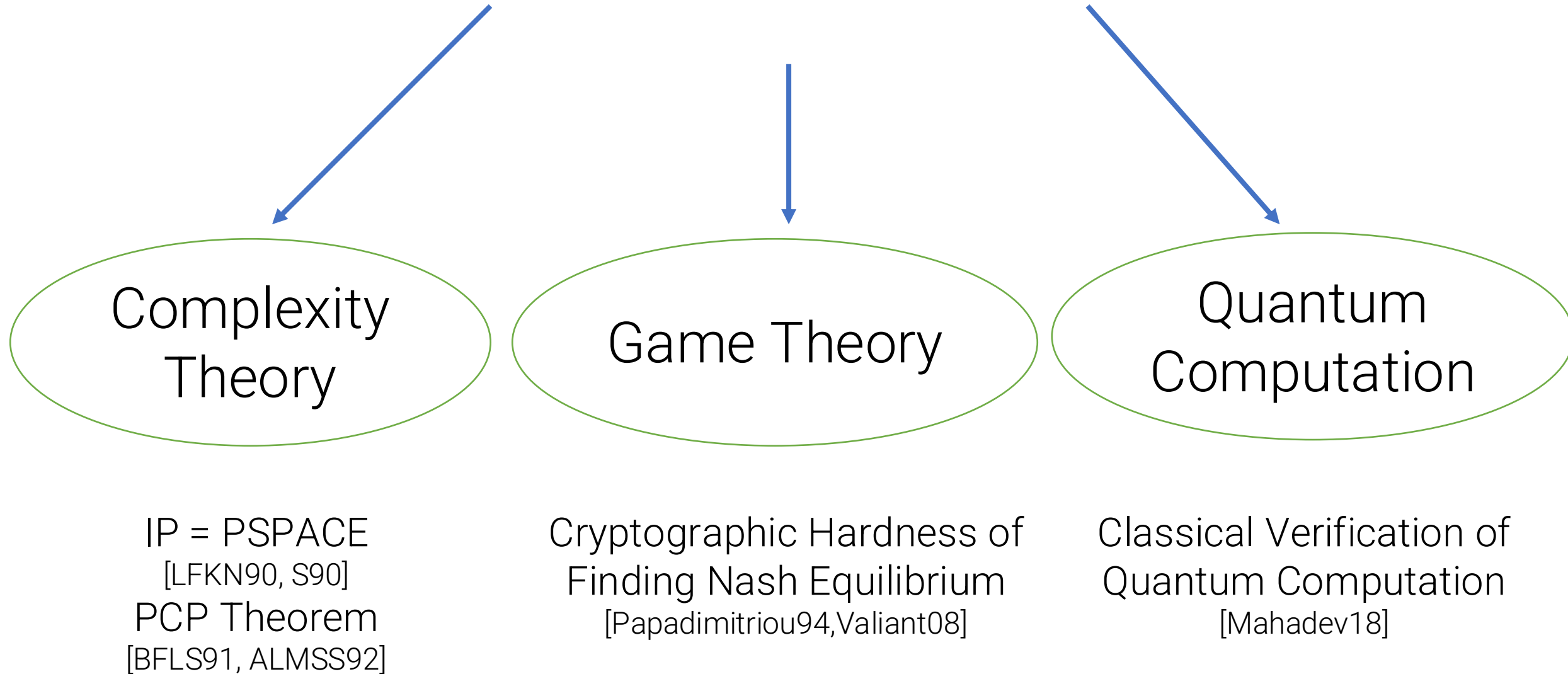
blockchains and cryptocurrencies

secure multiparty
computation



Cryptographic Proofs

Proofs: A Lens into Computation



Next time: Homomorphic Encryption